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EXAM – SOLUTION

Instructions:

- × *This is a 2h30 hours exam.*
- × ***BE SURE TO PUT THE STICKER ON THE EXAM.***
- × *There are a total of 100 points on the test + ~~15~~¹⁰ bonus points.*
- × *You have to answer all the questions.*
- × ***Use ONLY the space provided for answering the questions.***
- × ***You must write legibly.***

PART I - THE PRICE OF RISK (25 points + 10 bonus points)

Consider the following stylized representation of this L3 Macro class:

- × During the fall semester, students do 4 problem sets. Unlike you guys, they only have those 4 psets; no midterm, no final exam.
- × There are two different grading rules, rule *I* and rule *II*. Under rule *I*, all 4 psets are graded, and the final grade is the average of the 4 grades. Under rule *II*, only one of the 4 psets is graded; this graded pset is chosen at random; the grade received for the unique graded pset is the final grade. Formally, if the grades for each 4 psets are g_1, g_2, g_3 and g_4 , rule *I* gives grade g_I and rule *II* gives grade g_{II} defined by:

$$\text{Rule I : } g_I = \frac{1}{4} (g_1 + g_2 + g_3 + g_4)$$

$$\text{Rule II : } g_{II} = g_i \text{ with } i = 1, \dots, 4 \text{ with probability } \frac{1}{4}$$

- × All the students have exactly the same academic abilities. But they also all have good days and bad days. Half the time (good days), they do really well and get a 14/20 on their pset. The other half of the time (bad days), they don't do so good, and get a 6/20 on their pset. Good days and bad days are totally random (they happen with 50/50 chance each). Formally:

$$\forall i = 1, \dots, 4 : \begin{cases} \Pr[g_i = 6] &= \frac{1}{2} \\ \Pr[g_i = 14] &= \frac{1}{2} \end{cases}$$

- × All students care both about consumption (each of them consumes C euros each during the semester), and about their grades. All students have exactly the same “quasi linear mean-variance preferences” (don't worry, you are not supposed to know what that means). Formally, under each rule *I* and *II*, the utility they derive at the beginning of the semester, anticipating their future consumption C and their future grades g is given by,

$$\begin{aligned} U_I &= C + \alpha \mathbb{E}[g_I] - \beta \text{Var}[g_I] \\ U_{II} &= C + \alpha \mathbb{E}[g_{II}] - \beta \text{Var}[g_{II}] \end{aligned}$$

where for any random variable g , $\mathbb{E}[g]$ is the mean of g , and $\text{Var}[g]$ is the variance of g .¹

¹Reminder: if g is a discrete random variable such $g = g_1$ with probability p_1 and $g = g_2$ with probability p_2 such that $p_1 + p_2 = 1$,

$$\begin{aligned} \mathbb{E}[g] &= p_1 g_1 + p_2 g_2 \\ \text{Var}[g] &= p_1 (g_1 - \mathbb{E}[g])^2 + p_2 (g_2 - \mathbb{E}[g])^2 \end{aligned}$$

× We assume $\alpha > 0$ and $\beta > 0$.

× Assume for the moment that education is free, in the sense that students don't have to pay for their education themselves.

1 –

(a) Discuss what $\alpha > 0$ means (2 sentences max).

(b) Discuss what $\beta > 0$ means (2 sentences max).

(c) Explain what the parameters α and β correspond to respectively (4 sentences max).

(d) What would $\beta = 0$ correspond to? (2 sentences max).

- (a) $\alpha > 0$ simply means that students prefer higher grades than lower ones. If all grades are higher, $\mathbb{E}[g]$ increases, and the utility increases.
- (b) $\beta > 0$ means that students are risk averse. The variance of a random variable controls how much variation there is between realization, so that higher variance means more risk, and lower utility.
- (c) α controls the relative importance of grades compared to consumption. In other words, it “transforms” grades into § equivalents. β controls risk aversion. Again, it transform the variance (one among many potential measures of risk) into § equivalents.
- (d) $\beta = 0$ corresponds to risk neutrality. If $\beta = 0$, the student only cares about the average grade she expects, but does not care at all about the variance around this average.

2 –

(a) Calculate U_I and U_{II} , the utility derived from rule I versus rule II .

(Hint: for two independent random variables X and Y , $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$ and for any $\lambda \in \mathbb{R}$, $\text{Var}[\lambda X] = \lambda^2 \text{Var}[X]$).

Using $Var[g_i] = 16$ for any of the grades, i.e. any $i = 1, \dots, 4$, and $Var\left[\frac{1}{4}(g_1 + \dots + g_4)\right] = \left(\frac{1}{4}\right)^2 (Var[g_1] + \dots + Var[g_4]) = \frac{1}{16}(4 \times 16) = 4$, we have,

$$\begin{aligned} U_I &= C + 10\alpha - 4\beta \\ U_{II} &= C + 10\alpha - 16\beta \end{aligned}$$

3 –

- (a) Which of U_I and U_{II} is higher?
- (b) Give a brief explanation using simple words only (2 sentences max).

- (a) $U_I > U_{II}$.
- (b) Taking the average lowers the variance, in the sense that the variance of the average of 4 random variables is lower than the variance of each. Since students are risk averse, they prefer the less risky rule I than the more risky rule II .

× Now assume there are 200 students, and 2 separate macro classes, with 100 students in each class. Class I is graded according to rule I and class II is graded according to rule II .

× Education is no longer free. Students must pay 100 euros to join class I and a price p to join class II . C and p are denominated in euros.

× So the utility U_I derived from entering class I and the utility U_{II} derived from entering class II are given by:

$$\begin{aligned}U_I &= C - 100 + \alpha \mathbb{E}[g_I] - \beta \text{Var}[g_I] \\U_{II} &= C - p + \alpha \mathbb{E}[g_{II}] - \beta \text{Var}[g_{II}]\end{aligned}$$

× The University lets the market adjust prices.

× A market equilibrium is a price p such that $U_I = U_{II}$.

4 –

(a) What would happen if $U_I > U_{II}$? if $U_I < U_{II}$? (2+1=3 sentences max).

(b) Now explain why a price p such that $U_I = U_{II}$ corresponds to the notion of market equilibrium we have seen in class (2 sentences max).

(Hint: there are no first order conditions in that whole part; always use simple reasoning).

- (a) If $U_I > U_{II}$ everyone will prefer to take the class that uses rule I . If $U_I < U_{II}$, everyone will want to take the class with rule II . Since there are 100 seats in each class, if either $U_I > U_{II}$ or $U_I < U_{II}$, then some people are not going to get their preferred choice.
- (b) An equilibrium is a price such that *if* agents make optimal choices taking prices as given, *then* demand equal to supply in every market. If the price p is such that $U_I \neq U_{II}$, then from the previous question, demand cannot equal supply (demand is 200 for one class, 0 for the other, while supply is 100 for each class).

5 –

(a) Calculate the equilibrium price p .

(b) Is $p > 100$, $p < 100$ or $p = 100$?

(c) Explain why using only simple words (3 sentences max).

- (a) The equilibrium condition $U_I = U_{II}$ implies,

$$\begin{aligned} C - 100 + 10\alpha - 4\beta &= C - p + 10\alpha - 16\beta \\ \Rightarrow p &= 100 - 12\beta \end{aligned}$$

- (b) $p < 100$
- (c) Students are risk averse, and class II gives a more risky grade than class I . So if the price for both classes were the same, all students would prefer to go to class I , and there could not be an equilibrium. Making class II sufficiently cheaper is enough to make students indifferent between both classes, so that equilibrium between demand and supply can be achieved.

6 –

- (a) What is the equilibrium price if $\beta = 0$?
- (b) Is $p > 100$, $p < 100$ or $p = 100$ in that case?
- (c) Explain why using only simple words (2 sentences max).

- (a) $p = 100$.
- (b) $p = 100$.
- (c) if $\beta = 0$, then students are risk neutral, so that they are perfectly indifferent between taking class I or II . The prices for both classes must be equal so that students remain indifferent between both classes, and equilibrium between demand and supply can be achieved.

7 – Extra credit question (Warning: if you have answered all the question in that part perfectly, the extra credit questions should be relatively fast to answer; if you had a hard time answering the questions in that part, the extra credit questions will be very hard to answer; choose wisely whether or not you want to use your precious exam time on this extra credit)

× Assume now 100 students have preferences such that $\beta = 0$ and 100 students have preferences such that $\beta = \bar{\beta} > 0$.

- (a) What is the equilibrium price in this new setting with 2 types of students?
- (b) Is it unique?
- (c) Explain why using only simple words (3 sentences max).
- (d) Same three question with 99 students with $\beta = 0$ and 101 students with $\beta = \bar{\beta} > 0$.

- (a) Any price between $p = 100 - 12\bar{\beta}$ and $p = 100$ is an equilibrium.
- (b) No it is not unique, there actually are infinitely many equilibrium prices (if one can use fractions of cents that is).
- (c) The risk averse students would prefer class *II* for any price $p < 100$ and be indifferent if $p = 100$. The risk averse students ($\bar{\beta} > 0$) would prefer class *I* for any price $p > 100 - 12\bar{\beta}$, and be indifferent if $p = 100 - 12\bar{\beta}$. Since there are exactly 100 risk averse students (as many as there are seats in class *I*), and 100 risk neutral students (as many as there are seats in class *II*), any price between the lower bound $100 - 12\bar{\beta}$ and the upper bound 100 will be such that all risk averse students happily go to class *I* and all risk neutral happily go to class *II*, so that demand=supply for both classes.
- (d) In that case, there are too many risk averse students (or too few risk neutral students). The risk averse students are the marginal students deciding between class *I* and *II*, so the unique equilibrium price must be $p = 100 - 12\bar{\beta}$: this is the only way to make 100 of the risk averse students go to class *I*, and the 101st go to class *II*.

PART II - EASTER ISLAND (25 points)

This is an extract from “Easter Island: historical anecdote or warning for the future?”, by RAFAEL REUVENY and CHRISTOPHER S. DECKER, published in *Ecological Economics* 35 (2000) 271–287.

“Concerns regarding the Earth’s population explosion and the pressure it places on natural resources have rekindled interest in the sustainability of economic prosperity, an issue that seems particularly relevant as we enter a new millennium. Brown and Flavin (1999) explain, “the key limits as we approach the twenty-first century are fresh water, forests, rangelands, oceanic fisheries, biological diversity, and the global atmosphere.” They go on to ask, “Will we recognize the world’s natural limits and adjust our economies accordingly, or will we proceed to expand our ecological footprint until it is too late to turn back?” These concerns can be traced back to the work of Malthus (1798), who argued that population growth would eventually lead to natural resource depletion, economic decline, starvation, violent conflict, and population decline.

Many scholars believe that the decline of Easter Island exemplifies Malthus’s predictions (Weiskel, 1989; Ponting, 1991; Keegan, 1993; Brown and Flavin, 1999). The case of Easter Island stands at the center of our paper. A thousand years ago a civilization thrived there; by the time Europeans arrived in 1722, it had essentially disappeared and no one really knows why. While scholars have puzzled over this for years, offering many interesting theories, it was not until recently that formal economic modeling was applied to the subject. Brander and Taylor (1998) attempted to solve the mystery by modeling the island’s economy as a predator – prey system of renewable resource use in which the human population, dependent on the island’s resources for survival, overexploited them. This ultimately resulted in a dramatic population decline, a characteristic of the so-called “Malthusian Trap.” Brander and Taylor assumed that island’s resources and institutional structure were not subject to change or reform over time. We build on their work by relaxing these two restrictions. We then address the relevance of our work to modern countries, particularly less developed countries (LDCs).

Two solutions to the “Malthusian Trap” are typically suggested in the literature on sustainable development. The first solution involves institutional reforms. In this view, overexploitation of natural resources results from ill-defined property rights. Assigning private ownership to a resource will limit its extraction. Another kind of institutional reform involves population management efforts to reduce birth rates. The second solution to the “Malthusian Trap” is technological progress. Natural resource depletion need not be a concern if technological progress improves resource yields, carrying capacity and harvesting efficiency. In a sense, technology can provide “substitute” for natural resource depletion (Solow, 1997; Stiglitz, 1997). However, this approach is not without critics. In particular, Georgescu-Roegen (1971), Daly (1997) argued that capital and technology ultimately cannot provide substitute for natural resources.

Standard economic growth models emphasize the role of technological progress in securing perpetual growth. Armed with technology, population issues became tertiary to the analysis. Consequently, in these models population growth is determined exogenously, and typically is assumed either to be constant or to grow exponentially. Recently, dynamic models featuring endogenous population growth have appeared. The studies by Prskawetz et al. (1994), Milik and Prskawetz (1996), Brander and Taylor (1998) are particularly relevant to our paper. In the first two studies, a natural resource harvested under open access is combined with labor to produce a final good, but the consumption side is not modeled. In the Brander and Taylor study, a consumption side is added to a similar, but simplified, analytical

framework. All of these studies yield a similar system of differential equations, which is then parameterized and simulated. The goal of Prskawetz et al. and Milik and Prskawetz is to illustrate the general mathematical properties of such a model. Therefore, their parameterization is largely without empirical underpinning. Brander and Taylor's motivation is to study Easter Island specifically and thus their parameterization is chosen for that case.

Given the relatively primitive nature of the Easter Island civilization, it seems plausible to ignore – as Brander and Taylor do – institutional reforms and technological progress in this case. Yet, as stated earlier, the possible solutions to the “Malthusian Trap” involve precisely these two elements. We can now restate the two goals of our paper more fully. [...] we wish to investigate two hypothetical questions regarding Easter Island. What could have happened on Easter Island if its technology had been progressing? Similarly, what could have happened if population management had been adopted on the island?

1 – Write a 100 words summary of this introduction

- × There is a growing concern about the physical and ecological limits to growth given the explosion of population and natural resources utilisation.
- × This concern goes back to MALTHUS.
- × Easter Island is the typical example of MALTHUS's predictions.
- × BRANDER and TAYLOR have developed a model that shows that Easter Island has been in a Malthusian Trap.
- × The literature on sustainable development consider two solutions to evade the Malthusian Trap.
- × The first is institutional reforms that well-define property rights on natural resources.
- × The second is technological progress that improves resource yields, carrying capacity and harvesting efficiency.
- × This study extend the BRANDER and TAYLOR model to technological progress and population management.

2 – What is a *Malthusian trap*?

- (a) The Malthusian trap corresponds to a regime of economic and population growth described by MALTHUS.
- (b) He argues that population growth generally expanded in times and in regions of plenty until the size of the population relative to the primary resources caused distress (analogy with a population of rabbits on an island: rain → lot of grass → lots of baby rabbits survive → more rabbits → not more grass per rabbit).
- (c) Two types of checks hold population within resource limits:
 - × positive checks, which raise the death rate : *hunger, disease and war*;
 - × preventive checks, which lower the birth rate : *abortion, birth control, prostitution, postponement of marriage and celibacy*.
- (d) In simple words, the functioning of the malthusian regime is as follows:
 - × humans struggled for life;
 - × technological progress was insignificant ;
 - × technological progress and land expansion caused an increase in the size of the population, not in the standards of living;
 - × income per capita stayed in the proximity of a subsistence level.

3 – Given your knowledge of the BRANDER and TAYLOR (1998) model, answer the first question of REUVENY and DECKER: *What could have happened on Easter Island if its technology had been progressing?*

- (a) BRANDER and TAYLOR (1998) model :
 - × there is an equation for the resource (“forest”) dynamics: the natural growth rate of the resource is first increasing and then decreasing with the size of the resource;
 - × on top of the natural growth rate, the resource can be decreased by human activity (cutting trees);
 - × there is a second equation that describes the evolution of population, which growth rate is an increasing function of resources;
 - × the equilibrium level of the resource utilisation (woodcutting) is determined by market forces: population have the choice between producing wood (which is mainly a proxy for food) or another good (statues in the case of Easter Island, but more generally clothes, houses, etc...);
 - × the missing market in the model is the market for wood: it is a free resource with no property right.
 - × BRANDER and TAYLOR show that typically three steady states can arise in this economy:
 - “unoccupied or virgin island”: Population is zero and forest has its maximum size,
 - “collapsed island”: Population and forest size are zero,
 - “sustainable island” : Population and forest size take a positive value.

- (b) If technology progresses, it means that one can produce more food with the same amount of trees cut. This is making the interior steady state (sustainable island) sustainable with a larger population only if the steady state size of the forest is not too large (formally half the carrying capacity).

4 – Given your knowledge of the BRANDER and TAYLOR (1998) model, answer the second question of REUVENY and DECKER: *What could have happened if population management had been adopted on the island?*

- (a) No need to give again the model description.
- (b) Population management means that for the same level of food, population grows at a lower rate.
- (c) Therefore, the interior steady state will have a larger forest size and a smaller population size.
- (d) Such a fertility control has been observed in some Polynesian islands (but not on Rapa Nui).

Consider the following economy

- × There is a constant number of economic agents $2N$ that live two periods. They are young in the first period of their life and old in the second one.
- × Generations are overlapping, so that there are always N young and N old in each period.
- × There is only one good in that economy, that can be either consumed or saved. Its price is normalized to one.
- × Preferences are given by $U = \alpha \log c_{yt} + (1 - \alpha) \log c_{ot+1}$, $\alpha \in]0, 1[$, where c_{yt} is the consumption of an individual of generation t when young and c_{ot+1} the consumption of an individual of generation t when old.
- × When young, individuals of generation t supply inelastically one unit of labor and receive a wages w_t . They use their income to consume and to invest in capital k_t , so that $c_{yt} + k_t = w_t$.
- × When old, individuals of generation t rent their capital to firms at rate z_{t+1} and consume that return. Capital fully depreciates, so that $c_{ot+1} = z_{t+1}k_t$

1 – For given prices w_t and z_{t+1} , derive the optimal individual consumption of a generation t individual, as well as his savings. Also derive aggregate consumption C_t and aggregate supply of capital K_{t-1} in period t .

(a) An individual of generation t solves

$$\max_{c_{yt}, k_t, c_{ot+1}} \alpha \log c_{yt} + (1 - \alpha) \log c_{ot+1},$$

subject to:

$$\begin{aligned} c_{yt} + k_t &= w_t, \\ c_{ot+1} &= z_{t+1}k_t. \end{aligned}$$

The two budget constraints collapse to an intertemporal one:

$$c_{yt} + \frac{c_{ot+1}}{z_{t+1}} = w_t.$$

Write the Lagrangian $\mathcal{L}_t = \alpha \log c_{yt} + (1 - \alpha) \log c_{ot+1} + \lambda_t \left(w_t - c_{yt} - \frac{c_{ot+1}}{z_{t+1}} \right)$ and derive first order conditions to obtain:

$$\frac{\alpha}{c_{yt}} = \lambda_t = \frac{1 - \alpha}{c_{ot+1}}.$$

Using the intertemporal budget constraint, one obtains

$$\begin{aligned} c_{yt} &= \alpha w_t, \\ c_{ot+1} &= (1 - \alpha) z_{t+1} w_t, \\ k_t &= (1 - \alpha) w_t. \end{aligned}$$

(b) As there are N young and N old in the economy, aggregate quantities are $C_t = N\alpha w_t + N(1 - \alpha)z_t w_{t-1}$ and $K_t = N(1 - \alpha)w_t$.

As of technology,

- × There is a representative firm that produces the unique good of the economy according to the following technology: $Y_t = A_t K_{t-1}^\beta L_t^{1-\beta}$, with $\beta \in]0, 1[$. Its price is normalized to one.
- × A_t is a technological variable which is taken as given by the firm.
- × For the time being, assume $A_t = A \forall t$.

2 – From profit maximisation and assuming perfect competition, derive the representative firm first order conditions.

The representative firm maximizes $\Pi_t = AK_{t-1}^\beta L_t^{1-\beta} - w_t L_t - z_t K_{t-1}$ by optimally choosing L_t and K_{t-1} . First order conditions are:

$$\begin{aligned} z_t &= \beta AK_{t-1}^{\beta-1} L_t^{1-\beta}, \\ w_t &= (1-\beta)AK_{t-1}^\beta L_t^{-\beta}. \end{aligned}$$

3 – Write the equilibrium conditions on the good, labor and capital market in period t .

In equilibrium, the good market clears:

$$Nc_{ot} + Nc_{yt} = Y_t,$$

the labor market clears:

$$L_t = N,$$

and the capital market clears:

$$K_{t-1} = Nk_{t-1}.$$

4 – Using the labor and capital market equilibrium conditions, derive a difference equation for the aggregate capital stock in that economy. Plot K_t as a function of K_{t-1} and illustrates the dynamics of the economy. Is the economy growing in the long run? What is the limit of capital as time goes to infinity? Why?

- (a) The capital market equilibrium writes

$$K_t = (1 - \alpha)w_t N. \quad (\star)$$

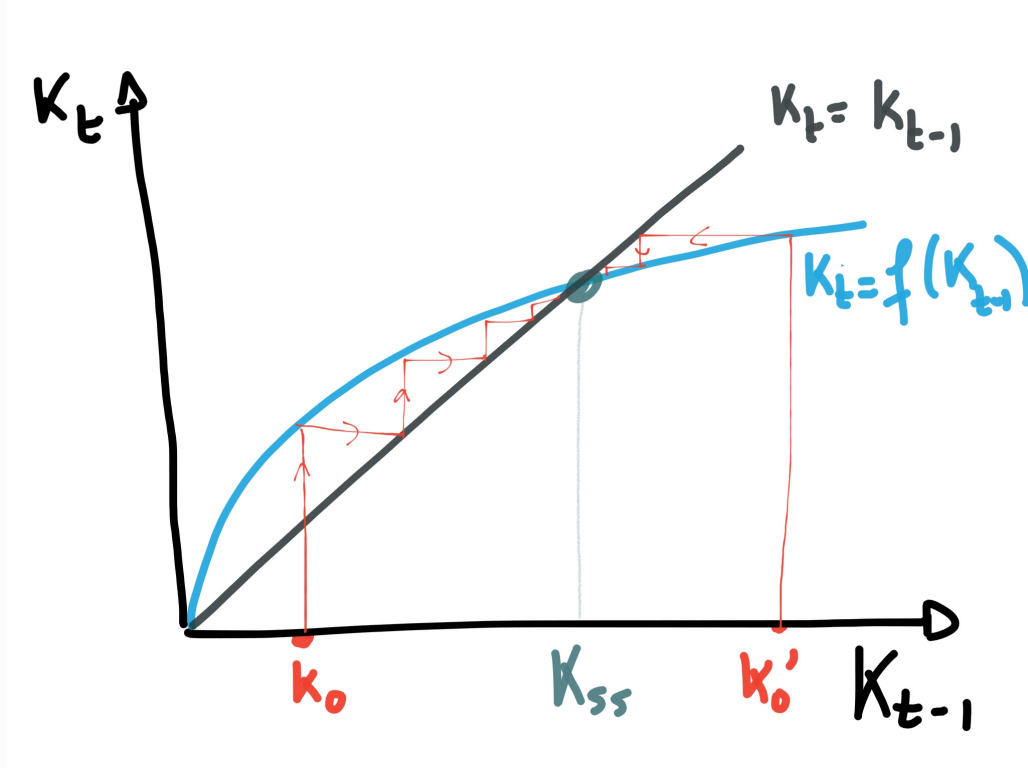
From the firm FOC, we have

$$\begin{aligned} w_t &= (1 - \beta)AK_{t-1}^\beta L_t^{-\beta} \\ &= (1 - \beta)AK_{t-1}^\beta N^{-\beta}, \end{aligned}$$

where we have used the labor market equilibrium condition. Use this last equation to replace w_t in $(\star)$ to obtain the difference equation

$$K_t = (1 - \alpha)(1 - \beta)AN^{1-\beta}K_{t-1}^\beta = f(K_{t-1}).$$

- (b)



- (c) The steady state of that economy is $K_{ss} = ((1 - \alpha)(1 - \beta)A)^{1/\beta} N$.
- (d) In the long run, the economy converges to the steady state and there is no growth. The limit of K_t is K_{ss} .
- (e) There is no growth because as capital is getting bigger and bigger, its marginal productivity is going down. At some point, it becomes so low that the amount of saving becomes exactly what is needed to replace the capital, but no more. At this point, the economy is at a steady state.

We now assume that A_t is not anymore a constant.

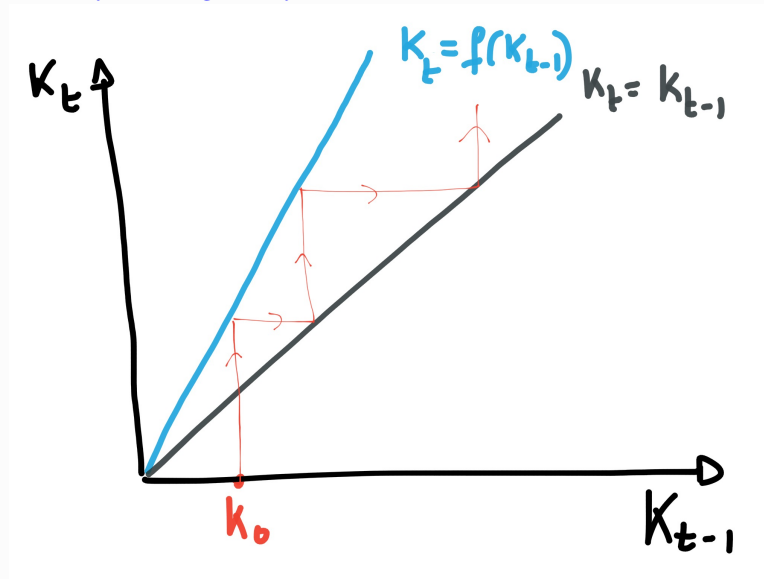
- × More specifically, we assume that aggregate capital acts as a productive externality, such that $A_t = \bar{A}K_{t-1}^{1-\beta}$, with the assumption $\bar{A} > ((1 - \alpha)(1 - \beta)N^{1-\beta})^{-1}$.

5 – What justification could you give for the assumption $A_t = \bar{A}K_{t-1}^{1-\beta}$?

- (a) Literally, this assumption means that factors are more productive when the existing stock of capital is higher.
- (b) This is an externality that can be explained by *Learning-By-Doing*. If the economy has produced and accumulated a lot in the past, K_{t-1} is higher and it is likely that the economy will be more productive today (high A_t).

6 – Using the new expression for A_t , find a new difference equation for K_t . Plot K_t as a function of K_{t-1} and illustrates the dynamics of the economy. Is the economy growing in the long run? What is the limit of capital as time goes to infinity? Why?

- (a) Take again the difference equation : $K_t = (1 - \alpha)(1 - \beta)A_t N^{1-\beta} K_{t-1}^\beta = f(K_{t-1})$,
and plug the expression for A_t to obtain : $K_t = ((1 - \alpha)(1 - \beta)\bar{A}N^{1-\beta}) K_{t-1}$.
- (b) The dynamics of the economy is now given by



- (c) Capital grows without limit.
- (d) There is unbounded growth because the overall production function has constant returns with respect to capital, so that its marginal productivity does not decrease with capital accumulation. Such a model is called an *endogenous growth* model.

PART IV - QUESTIONS (25 points)

1 – In the Cagan model of inflation, hyperinflation is always caused by expansionary monetary policy. True, False, Uncertain? (explain)

FALSE:

- (a) The CAGAN model is a monetary model that has been designed to understand the causal relationships between money demand, money supply and expectations.
- (b) The model consist of a money demand, a money supply and a demand = supply equation.
- (c) Money demand is a demand for real balances, is increasing in real income and is decreasing in the nominal interest rate.
- (d) By the Fisher equation, the nominal interest is equal to the real interest rate plus expected inflation.
- (e) The result of the model is that it is possible for inflation to become larger and larger with time *even if money supply is constant*, because expectations of future inflation feed-in into current inflation.
- (f) CAGAN have estimated his model on various periods of hyperinflation and has found that the German episode of the early 20's is indeed one in which inflation expectations did indeed contributed to hyperinflation.

2 – The three stages of economic growth

There are three stages of economic growth in history:

- (a) Malthusian Regime (before 1760): income per capita is roughly constant, close to subsistence level. Changes in the availability of resources translate into population changes (already discussed before)
- (b) Industrial Revolution and the demographic transition (1760-1900): rapid economic growth fueled by increasing production efficiency + increase of population smaller than the increase of total income.
- (c) The Great Divergence (after 1900): some countries became unprecedentedly rich while standards of living declined or grew slowly in other.

